EPFL CS-411 DIGITAL EDUCATION, MILESTONE 4

Docendo

Research plan: learning by teaching an AI classmate, with and without watching it use what you taught

Team Andrii Negrub, Phil Gustke, Rafael Cruz

Date 5 November 2026

Study docendo-jade.vercel.app
Team code `bandits2026` : both conditions, with a skip button

Code github.com/0LuminaX0/docendo

1 Research questions

Okita and Schwartz (2013, Experiment 2) found that students who taught a computer agent and then *watched it solve problems* with what they had taught learned more than students who solved the same problems themselves with the same right and wrong marks (d = 0.75): *recursive feedback*, feedback on one's own teaching seen through the pupil's actions. We test whether it holds with university students and a large language model as the pupil.

MAIN QUESTION (LEARNING OUTCOME)

Do students who watch an AI classmate solve practice problems from what they taught it, and correct it, score higher on a final test than students who solve the same problems themselves with right and wrong feedback? **Hypothesis:** yes.

SECONDARY QUESTION (LEARNING PROCESS)

How do learners teach and correct the AI classmate, and how do the two groups experience practice: how well they spot mistakes, what they say when correcting, and where they struggle?

The topic is multi-armed bandits: repeated choices between options with unknown average payoffs. The **learning goal** is to predict which option two decision rules, greedy and UCB, pick next and explain why: greedy can lock onto a worse option after one unlucky try, while UCB's bonus keeps it trying options it knows little about.

2 Participants and recruitment

Target population. University students of computer science or another engineering field, 18 or older, who have passed a course on probability or statistics and have *not* learned about bandits or reinforcement learning. Two screening questions at the start flag anyone who has; they are left out of the analysis.

Recruitment. We aim for 20 participants, 10 per condition, from our own networks, asked directly. We check the criteria orally and add those who agree to a WhatsApp group that receives the information sheet (attached). Each gets a one-use access code and does the study online between 9 and 15 November, told nothing about the conditions; pilot participants and anyone who knows the design are not recruited. Two pilots have run; about seven people have agreed or are being asked.

3 Study conditions and procedure

One website session of about an hour, the same for everyone except practice, the manipulated variable: the **AI group** practises with the AI classmate (recursive feedback), the **control group** without AI (direct feedback). Both groups teach the AI classmate first, as in the original study, so a difference comes from recursive feedback, not from teaching an AI.

STEP	TIME	WHAT THE PARTICIPANT DOES
0 Pretest	8 min	Two screening questions and five items on the maths and statistical reasoning the lesson builds on, with "I don't know".

STEP	TIME	WHAT THE PARTICIPANT DOES
1 Read	12 min	A text lesson in three short chapters (the setup, greedy, UCB) with worked figures, one chapter at a time. It closes before teaching and cannot be copied.
2 Teach Kai	16 min	Chat (typed or spoken) with Kai, an AI classmate who missed the lesson. Kai asks the same nine questions in the same words for everyone and writes down what it understood in its notebook, in the learner's words.
3 Practice	12 min	Three problems. **AI group:** Kai solves each problem step by step from its notebook; the learner says whether Kai is right, sees the mark, and after a wrong answer writes one correction, which Kai adds to its notes before trying again. **Control group:** the learner solves each problem, writing down the working, with a right or wrong mark and one retry.
4 Final test	12 min	Five multiple choice questions in new situations, each with a short written reason, which discourages guessing.
5 Questions	≈ 5 min	Six ratings and three open questions (Section 4); then the score and worked solutions.

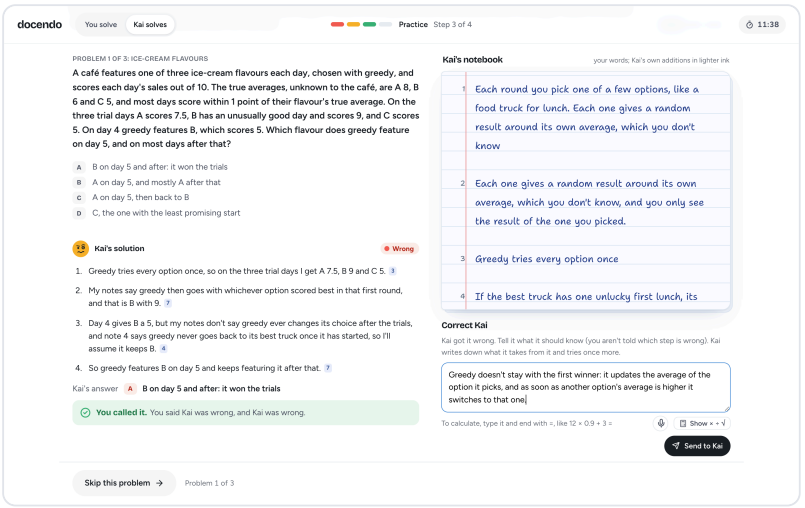


Figure 1. Practice in the AI group, with the live model. The learner taught that greedy "goes with whichever one scored best in the first round"; Kai follows that note (step 2), misses that greedy switches when another average overtakes, and answers wrongly. The learner called it and now corrects Kai.

How Kai records what it is taught. After each message a judge model checks it against a fixed list of the lesson's facts (nine that Kai waits for): which facts it explains correctly, partly or wrongly, and any wrong belief, each with a quote of the learner's words. Code checks every verdict: one whose quote is not in the message is dropped; a fact credited without its key point (e.g. that the bonus grows ever more slowly) counts as partial; a wrong belief is kept only if it is about the fact it is said to contradict. With simulated learners of fixed knowledge, the judge credited 98 of 98 facts they knew, none they did not, and caught 15 of 16 wrong beliefs.

Knowledge leakage checks. Kai must know only what it was taught, so the models are wrapped in several safeguards. The lesson is a graph of facts, and Kai's knowledge is only the part of it the learner has explained: the models that write Kai's replies and solve as Kai see Kai's notebook, never the lesson, its facts or the answer keys. Nothing Kai writes reaches the learner unchecked: a verification step checks every reply and solving step for lesson terms the learner has not used, and code, not the model, decides from the notebook whether Kai's answer may be right. Learners cannot hand Kai the lesson either: it closes before teaching, cannot be copied, and a message repeating it word for word is not written down. We tested these safeguards with simulated learners and with notebooks missing facts, and logged every check.

How Kai solves. A model told to "use only your notes" still uses what it knows (Do, Sonkar & Sachan, 2026), so code decides when Kai can be right: only when its notebook holds every fact the problem needs, correctly. A missing or wrong fact makes Kai reason from that gap, so its mistake comes from what it was taught. If Kai would solve every problem, it misreads one point its notes leave open (never one they state), defined in advance per problem and logged, so every learner gets to correct Kai at least once.

Assignment. Each participant's access code sets their condition, in balanced blocks of four. Two versions of the final test (A and B, different questions on the same ideas) and of the pretest are balanced within each condition, so every combination is equally frequent. All times are strict: a step ends at its limit.

4 Measures and analysis

QUESTION	DATA	ANALYSIS
Main: outcome	Final test score (0 to 5). Every wrong option is a known misconception; "I don't know" counts as wrong	AI vs control: difference in mean score as an effect size (Cohen's *d*, with a 95% interval) and a *t*-test; the same adjusted for pretest score (ANCOVA); learning gain from pretest to final test (as share correct)
Main: depth	The written reasons, scored 0 to 2 against a rubric	Scored blind to condition by two of us; same comparison as above
Secondary: process	Facts Kai understood after teaching; time per step; in the AI group, whether learners judged Kai's answers right and whether one correction fixed Kai	Descriptive, by condition; Kai's understanding checks that both groups taught equally well
Secondary: experience	Ratings (1 to 7) of effort, responsibility for Kai, understanding, practice's help and enjoyment; the manipulation check (whose understanding the marks were about)	Means by condition; perceived learning compared with measured scores
Qualitative	Three open questions (what helped, where it was difficult, how they decided after a wrong answer) and the AI group's corrections	A codebook from the pilot, applied by two of us independently (agreement as Cohen's κ); frequencies by condition, with examples

Power and uncertainty. Even if the original effect (d = 0.75) holds, a study with 10 per group would find it significant only about one time in three (power 0.36), so we report effect sizes with their intervals rather than rely on significance. Screened-out and broken-off sessions are excluded and reported.

5 Refinements since M3, and what the pilot changed

Besides the team's own changes (a text instead of videos, fixed questions for Kai), two pilot participants and several team runs found both test forms near 50% correct, as intended, and these problems:

FOUND	CHANGE
A session took 70 to 100 minutes	ε-greedy dropped from the lesson; three practice problems (from six) and five test questions (from ten); times set from the pilot
The text lesson read as thin, its figures as too simple	Rewritten: worked numbers in every figure, a passage per question of Kai's
Kai's questions missed facts practice needs, and repeated themselves	Nine fixed questions built on the lesson's facts, the same for everyone, skipping what was already taught
Kai solved everything for a learner who taught well, so practice taught nothing	A problem on the misconception both pilots showed (greedy keeps its first winner), and Kai's misreading of one open point (Section 3)

Still open. Some choices are not final: the time for each step, and the number of practice problems and final test questions (and which ones). The two pilots were run to check them, and we are adjusting them from the pilot data; once they are fixed, perhaps after one more short pilot, the main study runs, planned for 9 to 15 November.

References

Do, H., Sonkar, S., & Sachan, M. (2026). Simulating students or sycophantic problem solving? On misconception faithfulness of LLM simulators. arXiv:2605.12748.

Okita, S. Y., & Schwartz, D. L. (2013). Learning by teaching human pupils and teachable agents: The importance of recursive feedback. *J. of the Learning Sciences, 22*(3), 375 to 412.

Use of AI. Kai runs on Gemini 3.5 Flash Lite (questions and notes) and Claude Haiku 5.5 (solving). Claude Code helped build the website and draft this plan; the design decisions are ours. Each member's AI logs are in the submission.